

Questions for Engineering Physics tutorial test (PYM22)

Test-1

1. What is a cantilever? Derive the expression for Young's modulus of a cantilever loaded at the free end. [8]
2. Setup and solve the differential equation for Damped vibrations and discuss the case of over damping. [7]
3. An iron wire of length 1 m and radius 0.5 mm elongates by 0.25 mm when stretched by a force of 50 N, and twists through 0.6 radian when a torque of 4×10^{-3} Nm is applied to its ends. Calculate the Young's and Rigidity modulus of iron. [5]

Test-2

1. Solve Schrödinger wave equation for a particle in one-dimensional infinite potential well of width L to obtain energy Eigen values and Eigen functions. [8]
2. State and explain Heisenberg's uncertainty principle. Using this principle, show that an electron cannot be confined within the nucleus. [7]
3. The phase velocity of ripples on water surface is $v_p = \sqrt{2\pi s / \lambda \rho}$ where $s = 72 \text{ mNm}^{-1}$ is the surface tension and $\rho = 1000 \text{ kgm}^{-3}$ is the density of water. Find the group velocity, if the wavelength (λ) is 12cm. [5]

Test-3

1. Derive expression for thermo emf in terms of T_1 and T_2 . [8]
2. Describe the construction and working thermoelectric generator (TEG). [7]
3. Calculate the Neutral temperature for an iron silver thermo-couple. The values of a and b are 16.65 and -0.096 for iron and for silver these values are 2.86 and 0.017 respectively. [5]

Test-4

1. Describe the Cascade process for liquefying oxygen with a neat diagram. [8]
2. Explain Joule-Thomson effect. Derive an expression for Joule-Thomson cooling. [7]
3. The resistance of platinum wire thermometer is found to be 5.5Ω and 7.624Ω at 0°C and 100°C respectively. The resistance of the wire is found to be 14.45Ω when brought in contact with a hot body. Calculate the temperature of the body.

Unit-5

Important questions

1. Derive Bragg's law. Explain the Principle, construction and working of X-ray Diffractometer.
2. What is nanotechnology?. Explain quantum confinement and importance of surface to volume ratio.
3. With a neat diagram explain the Principle and working of AFM.
4. With neat diagram explain the Principle and working of Electron microscope.
5. Write a note on SEM and XPS.